

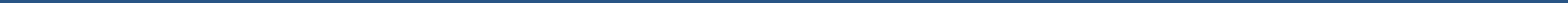
Bactericide Efficacy—Biocontrols And Natural Products

Bactericide	Resistance risk (FRAC group number) ^{1,5}	Fire blight ⁴		Phytotoxicity ⁷	Plant growth regulator/SAR
		contact	systemic		
Actinovate	low (BM 02)	2/3	0	1	0
AgriPhage-Fireblight	low (BM 02)	2/3	0	0	0
BacStop-OPL, -XL	low (NC—sanitizer)	2	0	1	0
Blight-Ban	low (BM 02)	2/3	0	1	0
Blossom Protect	low (BM 02)	4	0	1	0
Cinnerate, Seican, Cinnacure	low (BM 03)	2/3	0	0	0
Copper ²	low (M 01)	3	0	4	0
Dart	low (NC)	2/3	0	0	0
Double Nickel 55	low (BM 02)	2/3	0	1	0
LifeGard	low (P 06)	2	2	0	2
Lime sulfur/sulfur ³	low (M 02)	2	0	4	0
Regalia	low (P 05)	2/3	0	1	2
Sanitizers ⁶	low	2/3	0	0	0
Serenade	low (BM 02)	2/3	0	1	0

Rating: 5 = excellent and consistent, 4 = good and reliable, 3 = moderate and variable, 2 = limited and/or erratic, 1 = minimal and often ineffective, 0 = ineffective.

**	Not registered, label withdrawn or inactive in California.
1	Group numbers are assigned by the Fungicide Resistance Action Committee(http://www.frac.info/) (FRAC) according to different modes of action (for more information, see www.frac.info). NC = not coded in FRAC. Fungicides with different group numbers are suitable to alternate in a resistance management program. In California, make no more than one application of a fungicide with a mode-of-action group number associated with high resistance risk before rotating to a fungicide with a different mode-of-action group number; for other fungicides, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action group number.
2	Though copper may be effective for scab and blight control under low disease pressure, copper products may cause fruit scarring or russetting.
3	"Burns out" scab twig lesions when applied at delayed dormant and disrupts pseudothecial (or ascostroma) development when applied to leaves in fall. CAUTION: Lime Sulfur is incompatible with most other pesticides when used after budbreak. Check before use. Sulfur products are also used to thin flowers because of their phytotoxicity and indirectly control fire blight.
4	Plant growth regulators (PGR) such as prohexadione calcium (Apogee) can be used in an integrated approach to reduce host susceptibility but do not have direct activity against fire blight.

5	Biologicals: FRAC groups are provided as BM-No. or P-No. groups. In general, sulfur compounds are fungicidal and may affect applications of fungal biocontrols (e.g., Blossom Protect); whereas copper may affect applications of bacterial biocontrols (e.g., Actinovate, Bloomtime Biological, Blight Ban, Double Nickel 55, and Serenade). Rotations must consider these factors.
6	Sanitizers such peroxyacetic acid (e.g., Oxidate 5.0, ZeroTol HC, Perasan A) are oxidizers that act immediately on contact. They are neutralized rapidly by reducing agents and are non-persistent. Perasan A is for postharvest use in sanitizing fruit only.
7	Higher numbers indicate higher phytotoxicity.



Acknowledgment: Adaskaveg et al., 2025. [*Fungicides, Bactericides, Biocontrols, and Natural Products for Deciduous Tree Fruit and Nut, Citrus, Strawberry, and Vine Crops in California*\(PDF/PMG/fungicideefficacytiming.pdf\)](#).



UC IPM Pest Management Guidelines: Apple

UC ANR Publication 3432

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